Exercise Week 4

Advanced Applied Statistics

Fall 2025

Problem 1*

Let us assume that the datum x is generated by the Poisson density with an unknown parameter μ. Assume that the prior density for the parameter is $p(\mu) = e^{-\mu}$. Derive the expression for the the posterior density $p(\mu|x)$.

Problem 2*

Let the discrete random variable X be binomially distributed, that is, $X \sim \text{Bin}(n, \pi)$.

(a) Compute the Fisher information of the parameter π.

(b) Write down the Jeffreys prior for π.

(c) Write down the posterior density for the parameter π by assuming Jeffrey's prior.

Problem 3

A small café wants to estimate the average number of customers arriving per hour. During one randomly chosen hour, they observe $x = 4$ customers. Let the observation $X \sim \text{Poisson}(\theta)$, where θ is the true arrival rate per hour. Assume the prior $\theta \sim \text{Gamma}(\alpha = 2, \beta = 1)$, where β is the rate parameter (so the prior mean is α/β).

(a) Write down the likelihood function.

(b) Derive the posterior distribution for θ.

(c) Identify the posterior as a Gamma distribution (state its parameters).

(d) Compute the posterior mean of θ.

(e) Compare the prior mean and posterior mean. How did the data change your belief about the café's customer rate?

Problem 4

Three students take a short quiz with only 2 questions each. The students performed as folows

- Student A: 2/2
- Student B: 0/2
- Student C: 1/2

Assume each student has their own probability of answering correctly, θ_i. Further assume the likelihood $X_i \sim \text{Binomial}(2, \theta_i)$ and prior $\theta_i \sim \text{Beta}(\alpha, \beta)$.

(a) Compute the *raw proportions* (MLEs) of correct answers for each student.

(b) Suppose we use an *empirical Bayes prior* where the prior mean is set equal to the overall average proportion of correct answers across all students. (Hint: The overall mean is $(2 + 0 + 1)/(2 + 2 + 2)$.)

(c) With this prior mean, derive the posterior mean for each student

$$\mathbb{E}[\theta_i \mid x_i] = \frac{x_i + \hat{\alpha}}{2 + \hat{\alpha} + \hat{\beta}}$$

(You do not need exact values of $\hat{\alpha}, \hat{\beta}$—just explain how the shrinkage works.)

(d) Compare the raw proportions vs. the empirical Bayes posterior means. Which student's estimate is pulled the most toward the overall average? Why?